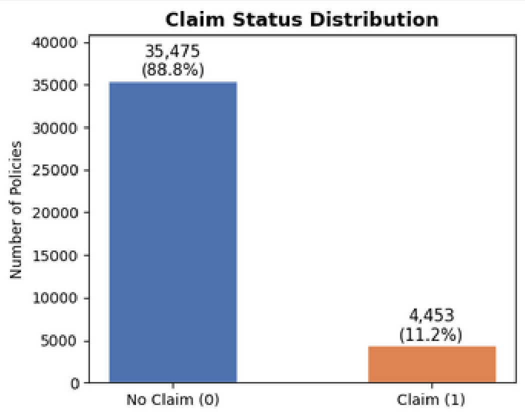
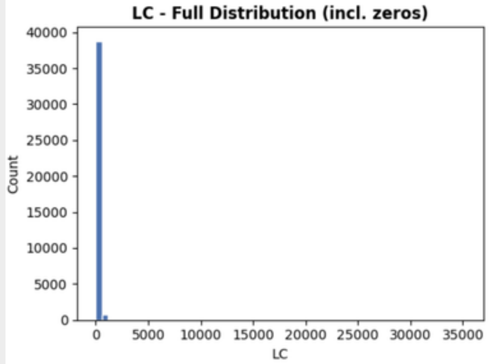
Insurance Risk Prediction: Claims & Loss

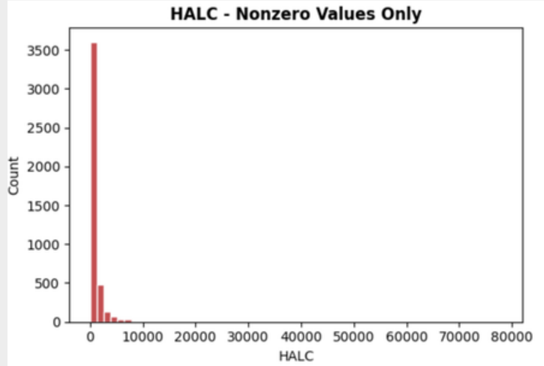
Problem & Data

Business Problem:

- Insurance pricing must reflect true risk
- Mispricing → adverse selection → financial loss







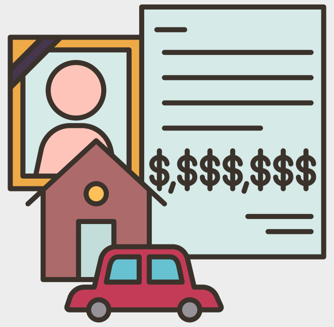
Data Challenges:

- Class imbalance (88.8% no claims)
- Highly skewed loss distribution (many zeros + extreme values)

Methodology

Task 1 (Loss Prediction):

- Tweedie GLM (baseline)
- Two-stage LightGBM (final)
- E(Loss) = P(claim) × E(severity | claim)



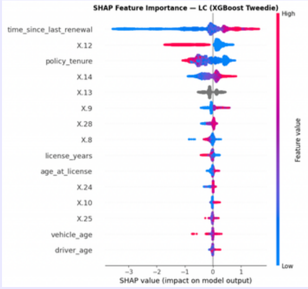
Task 2 (Claim Likelihood):

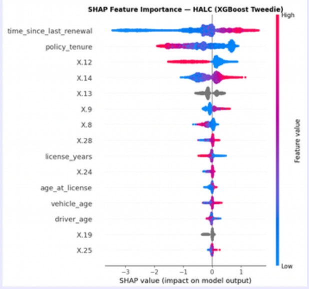
- Logistic Regression (baseline)
- Two-Stage LightGBM: Stage 1 classifier (final)

Model Performance

Loss Prediction (LS & HALC)

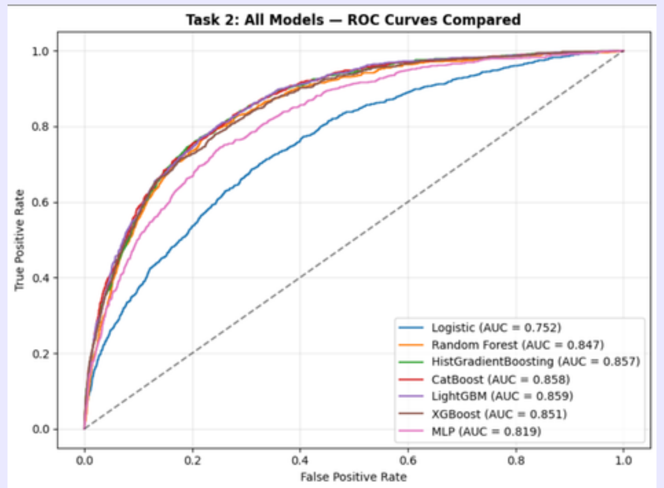
Model	LC MSE	HALC MSE
GLM Tweedie (statsmodels)	257,462	1,016,123
Ridge Tweedie GLM (sklearn, α=1.0)	255,530	1,014,256
XGBoost Tweedie (direct)	253,432	1,002,833
LightGBM Tweedie (direct)	253,491	999,292
Stacking (GLM + XGB + LGB)	251,614	994,860
Two-Stage XGBoost	251,211	992,119
Two-Stage LightGBM	**251,107**	**992,062**





- Tweedie regressor accounts for skew and large spikes of zeroes
- GLM models explains baseline interpretability with statistical significance, but cannot capture nonlinear interactions
- Random Forest and XG Boost (decision tree models) decrease LS MSE and capture nonlinear features
- 88.8% of policies have zero claims (split into classifier and regressor)

Claim Likelihood (CS)

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Confusion Matrix (threshold = 0.5):
[[6996   82]
 [ 744  164]]
              precision    recall  f1-score   support

    No Claim       0.90      0.99      0.94      7078
       Claim       0.67      0.18      0.28       908

    accuracy                           0.90      7986
   macro avg       0.79      0.58      0.61      7986
weighted avg       0.88      0.90      0.87      7986

Confusion Matrix (threshold = 0.3):
[[6671  407]
 [ 516  392]]
              precision    recall  f1-score   support

    No Claim       0.93      0.94      0.94      7078
       Claim       0.49      0.43      0.46       908

    accuracy                           0.88      7986
   macro avg       0.71      0.69      0.70      7986
weighted avg       0.88      0.88      0.88      7986
```

- Boosting models outperform linear baseline (AUC +13%)
- LightGBM and CatBoost achieve equivalent AUC (0.86). Unified framework adopted.
- Severe class imbalance leads to low recall at threshold 0.5 (≈ 18%)
- Lowering threshold to 0.3 improves recall (→ ≈ 43%), highlighting the importance of threshold tuning

Insights & Recommendations

Model Insights

- Nonlinear relationships and multicollinearity are critical for risk prediction
- Class imbalance and skewed data are the main modeling challenges
  - Few extreme observations and heavy skew for policies with zero claims
- Tune threshold for real-world decision-making
- Misclassification of high-risk customers is more costly than false alarms
- Models may not generalize across different insurance contexts (auto vs. property or personal vs. commercial)

Recommendations

- **Adopt frequency–severity decomposition for pricing:** modeling claim probability and severity separately yields more accurate premiums than aggregate prediction
- **Tune thresholds toward higher recall:** lowering from 0.5 to 0.3 increases claim detection from 18% to 43%, reducing costly underpricing of high-risk policyholders
- **Incorporate renewal timing and payment behavior into risk scoring:** these underutilized signals consistently rank among the top predictors alongside traditional factors
- **Consider incorporating telematic data**